Scalvini Florian

Postdoctoral researcher at IMT Atlantique, Brest

florian.scalvini@imt-atlantique.fr |(+33)6 45 57 04 38 | https://florianscalvini.github.io/

Research Experience

Postdoctoral Research Scientist 2024~2026

IMT Atlantique ~ LaTiM (UMR1101)

"Structural neonatal brain developmental trajectory"

As part of the ANR HINT project, this postdoctoral research investigates the estimation of cortical deformation during folding in the developing fetal and neonatal mammalian brain. The work focuses on diffeomorphic image registration frameworks to model the complex and rapid structural changes occurring during gyrification, with a particular emphasis on leveraging longitudinal multi-species MRI data to improve registration accuracy and robustness across early developmental stages.

Supervisor: François Rousseau &Nicolas Passat

Ph.D. in Computer Science 2020~2024

Université de Bourgogne ~ ImViA (EA7535)

" Methods and systems for assisting blind people based on auditory perception of visual scenes"

Conducted doctoral research at the intersection of computer vision, spatial audio, and assistive technology, developing real-time sensory substitution systems to support the navigation of visually impaired people. Work spanned the full pipeline from scene analysis and obstacle detection to 3D sound rendering and human-computer interaction, with a strong emphasis on low-latency, deployable solutions for both indoor and outdoor environments

Supervisors : Julien Dubois, Cyrille Migniot & Maxime Ambard

Education

Master's degree in Image and Artificial Intelligence 2019~2020

Université de Bourgogne

Keys : Geometric modelling, Image processing (Pytorch) & Medical image processing

Bachelor of Engineering 2017~2020

ESIREM (Polytech Dijon) ~ Université de Bourgogne

Specialization in Embedded Systems

Key: Real-time software development & Image processing

Experience

Software/Electronics Development Engineer 2017~2020
SATT Sayens

Complementary Teaching Assignments 2017~2020
ESIREM (Polytech Dijon) ~ Université de Bourgogne
Keys: Development in C++ and Python

Publication

Journals :
- Bordeau, **Scalvini**, Migniot, Argon, Dubois, Ambard: Localization abilities with a visual-to-auditory substitution device are modulated by the spatial arrangement of the scene, *Attention, Perception, & Psychophysics,* 2025
- **Scalvini**, Bordeau, Ambard, Migniot, Dubois; Outdoor Navigation Assistive System Based on Robust and Real-Time Visual–Auditory Substitution Approach, *Sensors*; 2024
- **Scalvini**, Bordeau, Ambard, Migniot, Vergnaud, Dubois; uB-VisioGeoloc: An image sequences dataset of pedestrian navigation, including geolocalised-inertial information and spatial sound rendering of the urban environment's obstacles; *Data In Brief*; 2024
- **Bordeau**, Scalvini, Migniot, Argon, Dubois, Ambard; Cross-modal correspondence enhances elevation localization in visual-to-auditory sensory substitution; *Frontiers in Psychology,* 2022

Conferences :
- **Scalvini**, Passat, Rousseau : Analyse du développement cérébral précoce par recalage d'IRM longitudinales basée sur un champ de vélocité stationnaire, *IABM,* 2026*,* Lyon
- **Scalvini**, Passat, Rousseau : Estimation des déformations longitudinales cérébrales basée sur un champ de vélocité stationnaire, *IABM*, 2025, Nice
- **Scalvini**, Bordeau, Ambard, Migniot, Argon, Dubois; Low-Latency Human-Computer Auditory Interface Based on Real-Time Vision Analysis; ICASSP; 2022, Singapour
- Scalvini, **Bordeau**, Ambard, Migniot, Argon, Dubois; Visual-auditory substitution device for indoor navigation based on fast visual marker detection; *SITIS*; 2022, Dijon
- **Bordeau**, Scalvini, Migniot, Argon, Dubois, Ambard; Distance perception of objects using visual-to-auditory sensory substitution: comparison of conversion methods based on sound intensity and envelope modulation; *APCAM*; 2022, Boston

Referees

- Prof. François Rousseau, IMT Atlantique, LaTiM, francois.rousseau@imt-atlantique.fr
- Prof. Julien Dubois, Université de Bourgogne, ImViA, julien.dubois@u-bourgogne.fr
- Dr. Cyrille Mignot, Université de Bourgogne, ImViA, cyrille.migniot@u-bourgogne.fr